

Adib BEN HELAL

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ABOUT ME

Master Data Science student with a background in software engineering, passionate about building data-driven solutions. Experienced in end-to-end data pipelines, machine learning, and computer vision through hands-on academic projects. Currently Seeking an opportunity as an **AI & Data Scientist Engineer** to leverage my technical skills, contribute to innovative projects, and develop my expertise within a dynamic team.

WORK EXPERIENCE

Triweb - Tunisia, Charguia 2 Tunis

Email: contact@tri-web.com | **Website:** https://tri-web.com/ | **Link:** https://drive.google.com/file/d/11FLb4k_qOEDEgmugj7ynWSiexszHF-It/view?usp=drive_link

Final Year Project Internship (PFE)

[01/02/2025 - 01/06/2025]

Role: Developed an AI-based recruitment platform integrating semantic CV analysis. Designed a job-matching model and a personalized quiz generation system using **Gemini 2.0 Flash**. Applied **NLP** techniques for semantic analysis and **YOLOv8** for CV sections extraction. Built the web application using the **MERN stack**, and implemented the intelligent agent with **Python** and**Flask API**.

DevStage TEL: +33 750 53 30 31 - Tunisia, El MEY DJERBA

Part-Time Job

[01/11/2024 - 02/12/2024]

Role: Developed the backend for a Djerba Student Trip e-ticket reservation platform using **MongoDB, Express, and Node**.

DigitalCook - Tunisia, HOUMT-SOUK

Website: https://digitalcook.tn/

Web developer : Summer Internship

[01/06/2024 - 02/07/2024]

Role: Developed and deployed the frontend for a Djerba Student Trip e-ticket reservation website using **React** and **TypeScript**. Designed a responsive interface with seat selection, booking, and seamless data interactions.

djerbainternationale agency - Tunisia, HOUMT-SOUK Djerba

Website: https://djerbainternational.com/

Web developer : Summer Internship

[01/06/2023 - 02/07/2023]

Role: Built an informative webpage with **WordPress** about eco-friendly bike trips in Djerba.

EDUCATION & TRAINING

Graph RAG with LangChain and Neo4j

datacamp [01/08/2026 - 02/08/2026]
https://www.datacamp.com/completed/statement-of-accomplishment/course/8bc6620a617a903ad0095fb10492e0fdc647ead0?utm_medium=organic_social&utm_campaign=sharewidget&utm_content=soa

Data Science Master

Higher Institute of Informatics and Mathematics of Monastir [01/09/2025 - Current]

City: MONASTIR | **Country:** Tunisia

Website: http://www.isimm.rnu.tn/public/isimm

software engineering | bachelor's degree

Higher Institute of Informatics and Mathematics of Monastir [12/09/2022 - 25/06/2025]

City: MONASTIR | **Country:** Tunisia

Website: www.isimm.rnu.tn/public/isimm

experimental sciences | High-SchoolDiploma

Erriadh Hight school Djerba [15/09/2018 - 30/06/2022]

City: HOUMT-SOUK Djerba | **Country:** Tunisia

LANGUAGE SKILLS

Mother tongue(s): Arabic

French

LISTENING: B2 **READING:** B2 **WRITING:** B1
SPOKEN PRODUCTION: B1
SPOKEN INTERACTION: B1

English

LISTENING: B1 **READING:** B1 **WRITING:** A2
SPOKEN PRODUCTION: A2
SPOKEN INTERACTION: A2

deutsch

LISTENING: A1 **READING:** A1 **WRITING:** A1
SPOKEN PRODUCTION: A1
SPOKEN INTERACTION: A1

NETWORKS AND MEMBERSHIPS

[31/05/2024 - Current] Erriadh, Djerba,tunisia

Association Djerba Tawasel Active participation in cultural and social activities, helping to promote Djerba's unique architecture and traditions, in line with its recognition on the UNESCO World Heritage List.

Summer 2025 – Contributed to the organization of the 2nd Ibadi Youth Meeting in Djerba, including a field visit to historical landmarks of Djerba Island. Designed and managed the registration process using Microsoft Forms, and automated the sending of personalized e-tickets via Microsoft Power Automate.
https://www.facebook.com/jerbatawasolriadh

PROJECTS

[01/03/2026 - 31/03/2026]

Hydrolytics (Water Resource Management Pipeline) End-to-end data engineering pipeline for smart

agricultural irrigation, integrating real-time IoT sensor streaming via Apache Kafka, batch ingestion from Open-Meteo & NASA POWER APIs, and automated ETL orchestration with n8n. Computes evapotranspiration, hydric stress scores, and irrigation recommendations stored in PostgreSQL, visualized through a live Streamlit dashboard. Fully containerized with Docker Compose.

Links: https://drive.google.com/file/d/1YMt2jgGRc5Hm_1O9K72Zs_jiq48BCHCc/view?usp=drive_link | https://github.com/AdibBENHLEL/Hydrolytics

[01/04/2026 - 30/04/2026]

OR Shield (Surgical Mask Detection) Real-time computer vision application for operating room access control, built with a custom-trained YOLOv8n model (mAP50: 0.93) on 1,510 annotated images. Features a Flask REST API backend and React/Vite frontend with live webcam bounding-box annotations, confidence scoring, audio alerts, and a session statistics dashboard.

Links: https://drive.google.com/file/d/1N4MCZ-vF2qoa6uxknh07G4UW41Cu1IT3/view?usp=sharing | https://github.com/AdibBENHLEL/Mask_No_Mask_detection_YOLO_v8

[01/04/2026 - 15/04/2026]

Fraud Detection with Hadoop MapReduce Designed and deployed a distributed fraud detection pipeline on a multi-container Hadoop cluster (Docker). Implemented a MapReduce workflow in Python using Hadoop Streaming to process the PaySim financial dataset (450MB+), applying rule-based detection logic (transaction frequency, balance anomalies, transfer/cash-out patterns) to flag suspicious accounts at scale.

[01/12/2024 - 10/12/2024]

CNN Image Classification – Happy vs. Sad Faces Role: Developed a **convolutional neural network** using **TensorFlow** to classify facial expressions (happy vs. sad) from image datasets.

[10/10/2024 - 31/10/2024]

Mobile Stock Management App Role: Developed an Android mobile application for stock management using **Java, XML, and SQLite**.

[02/04/2024 - 14/04/2024]

DirennaPlan – Desktop To-Do List App Role: Created a desktop application for task management using **Python** and **Tkinter**.

[01/03/2024 - 15/03/2024]

Python Turtle Maze Solver (BFS, DFS, A*) Role: Developed an application to generate and solve mazes using the Turtle library and pathfinding algorithms such as **BFS, DFS, and A***.

[01/02/2024 - 15/02/2024]

E-learning Platform (PFA) Role: Designed an e-learning platform using **UML** diagrams (use cases, classes, sequences, activities).

[01/11/2023 - 15/11/2023]

Stock Management Desktop App Role: Developed a **Java**,**JDBC** and **SQL**-based desktop app for managing stock.

[01/10/2023 - 15/10/2023]

Simulation Data Visualisation data Analysis Desktop App Role: Created a **Python**(**Matplotlib**-**Tkinter**)desktop app to evaluate the attractiveness of new casino card games.

SKILLS

c | python | html | css | react js | php | java script | TypeScript | MySQL | SQL Server Management Studio | a2hosting | wordpress | Android Studio | cPanel | XAMPP | Odoo, | Docker, | java | Postman | Canva | FileZilla | Git | GitHub | Material-UI | Microsoft Power Automate | Power Pivot BI | Matplotlib | Tkinter | PyQt5 |

TensorFlow | NLP | MERN stack | VMware | VirtualBox | capcut | Flask API | YOLOv8 | Apache Kafka | ETL Pipelines | Batch & Stream Processing | PostgreSQL | MinIO | Pandas | n8n | Pytest | Hadoop | SPARK | Talend Open Studio for Data Integration (ETL) | D3 Js | Graph RAG | LangChain | Neo4j

HOBBIES AND INTERESTS

Video Editing and Content Creation https://youtube.com/@hdari6170?si=_yZL745cS3hgm3M1
https://youtube.com/channel/UCcvPzdqMKdKF1cgHNFyBVdA?si=YvNbASHOYEGi2Wo9
https://youtube.com/@adibbenhelal?si=ewuQlqpL1SAiJdhN

Cycling

Swimming

Fishing

Card games (Belote)